Yahye Bashir

Robotics Software Engineer | Robot Learning

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Summary

Graduate Robotics researcher building robot learning pipelines end to end: from data collection and simulation to policy training and evaluation. Experience with VLA Training and fine-tuning (SmolVLA, LeRobot), reinforcement learning (PPO, SAC) in Isaac Lab and MuJoCo, and classical manipulation control. developed end-to-end autonomy stack of the MORIA humanoid mobile manipulator in ROS 2 and isaacsim.

Education

Bursa Technical University 2025 – 2027 (expected)
M.Sc. Mechatronics Engineering (Robotics)
Karabuk University 2019 – 2024
B.Sc. Mechanical Engineering
Northwestern University (Coursera) — Modern Robotics Specialization: kinematics, dynamics, control, planning

Experience

Robotics Software Engineer — MORIA Humanoid Robot 04/2025 – Present
ROTASAM Research Center, Bursa Technical University, Bursa, Türkiye

- Develop the autonomy stack of MORIA, a mobile humanoid manipulator (dual 5-DOF arms, five-finger hands) for autonomous book retrieval in libraries; digital twins in Gazebo and Isaac Sim with ROS 2, MoveIt 2 and `ros2_control`.
- **Perception:** RGB-D book detection and pose estimation with point-cloud segmentation, ICP registration and deep instance segmentation.
- **Planning and grasping:** collision-aware motion planning (OMPL, trajectory optimization) and multi-finger grasp design.
- **Mobile manipulation:** reachability-based base placement, navigation to the shelf, and the full pick–transport–place cycle with MoveIt Task Constructor.

Robotics Software Engineer (Volunteer) 09/2024 – 03/2025
BTU Space, Aviation and Defence Industry Club, Bursa, Türkiye
Mechanical Engineering Intern 09/2022 – 10/2022
Drone Market, Sakarya, Türkiye
Control Engineering Intern 08/2022 – 10/2022
Karabuk Teknokent, Karabuk, Türkiye

Selected Projects

- **Egocentric Phone Video to a VLA Robot Policy** `ego2libero_smolvla`
  Recorded 48 head-mounted iPhone demonstrations and turned them into robot training data for a Franka Panda in LIBERO. Built perception , keypose retargeting from known contact heights, and closed-loop replay (96/96 successful). Fine-tuned SmolVLA (450M) with LeRobot: 28% success. Traced robot jitter to noise-injected labels (18× jerkier), which raised success from 12% to 28%.
- **RL Manipulation (UR10e + Robotiq)** — Reach–grasp–lift policies with RSL-RL and SKRL in Isaac Lab; multi-stage reward shaping; generalization across object poses; ROS 2/MoveIt 2 integration.
- **RL Quadruped Locomotion** — PPO walking policies in Isaac Lab for a custom quadruped; reward design for velocity tracking and gait stability; tested under domain randomization and rough terrain.
- **SO-101 Arm: Classical Control and RL** — From-scratch FK, IK, Jacobians, RNEA and trajectory tracking in MuJoCo; SAC pick-and-lift policy; MoveIt 2–Isaac Sim bridge with an RSL-RL

pick-and-place policy in Isaac Lab.

- **Model-Based Control in MuJoCo** — WidowX 250S: quintic/SLERP trajectories, numerical IK and computed-torque control (M, C, G) for pick-and-place. Unitree A1: gait scheduling, swing trajectories and stance-force control for trotting.
- **ROS 2 Systems** — Vision-guided colour sorting with a Franka Panda (MoveIt 2, OMPL); autonomous patrolling mobile robot (Navigation2, Gazebo).

Technical Skills

Robot learning: PyTorch, LeRobot, VLA's, imitation learning, RL (PPO, SAC), RSL-RL, SKRL
Simulation: MuJoCo, NVIDIA Isaac Sim, Isaac Lab, LIBERO, Gazebo
Robotics: ROS 2, MoveIt 2, `ros2_control`, Navigation2; kinematics, dynamics, trajectory optimization
Perception: OpenCV, MediaPipe, Open3D, PCL
Control: PID, computed-torque, LQR, MPC
Programming and tools: Python, C++, MATLAB, Linux, Git